Project Report: Credit Card Customer Prediction Analysis

1. Overview & Problem Statement

Acquiring a new banking customer is significantly more expensive than retaining an existing one. Currently, the bank is experiencing a customer churn rate of 16.07%. Traditional spreadsheet reporting fails to quickly highlight the overlapping factors that lead to account closures. This project solves that problem by transforming flat data into an interactive visual model that instantly reveals the shared traits of churning customers, allowing the retention team to act proactively.

2. Tools & Technologies

- **Data Cleaning & ETL:** Microsoft Excel (Handling nulls, standardizing categorical text, checking data integrity).
- **Data Modeling & Visualization:** Microsoft Power BI Desktop.
- **Calculation Engine:** DAX (Data Analysis Expressions).
- **Dataset Source:** Kaggle.

3. Methodology

Phase 1: Data Preprocessing (Excel)

- Cleaned raw Kaggle data, ensuring unique identifiers (`CLIENTNUM`) were intact and formatted correctly.
- Standardized text fields and evaluated "Unknown" entries in columns like `Income_Category` and `Education_Level` to ensure they did not skew the visualizations.
- Verified numerical data types for financial columns to ensure Power BI could accurately aggregate sums and averages.

Phase 2: Data Modeling & Visualization (Power BI)

- Imported the cleaned dataset and established the necessary data hierarchy.
- Designed a UI/UX layout focused on top-down analysis: high-level KPIs at the top, followed by categorical breakdowns, and detailed scatter plots at the bottom.
- Implemented interactive visuals including clustered bar charts, donut charts, and a complex scatter plot for financial behavior.

Phase 3: DAX Development

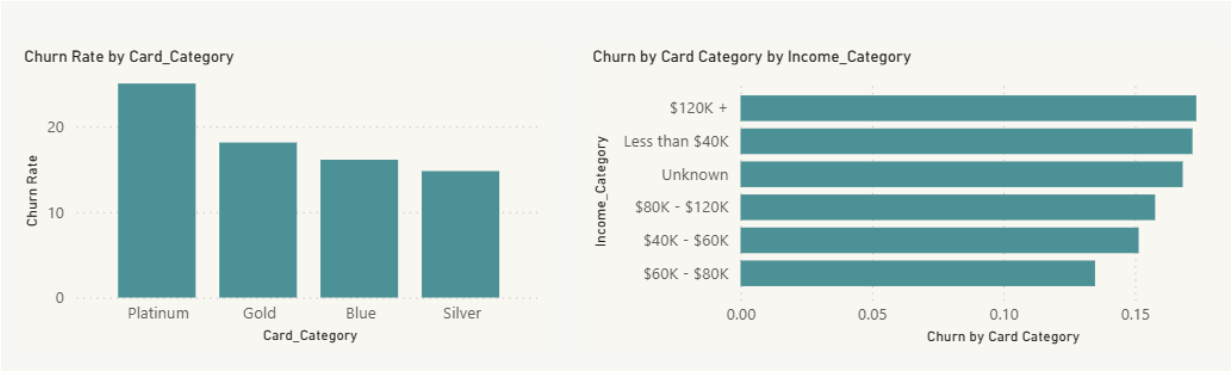
Created specific DAX measures to build dynamic KPIs, ensuring metrics recalculate accurately when dashboard filters are applied. Notable metrics derived include:

- **Churn Rate:** Calculated as `16.07%`.
- **Total Active Customers:** `8,500`.
- **Average Customer Age:** `46.33` years.
- **Sum of Average Utilization Ratio:** `2.78K`.

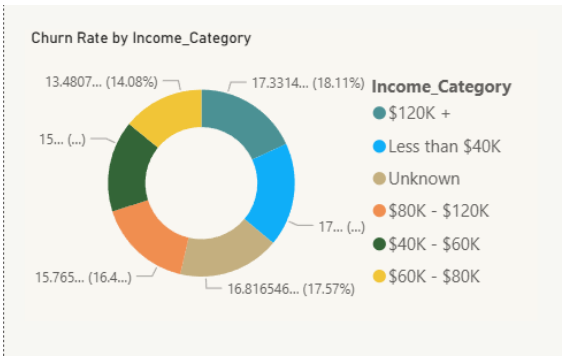
4. Key Insights & Business Findings

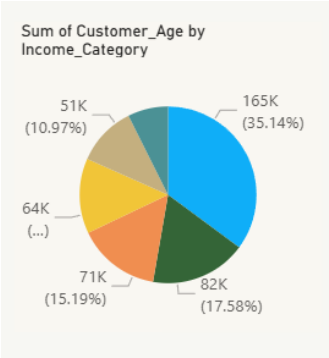
Based on the dashboard visualizations, several critical risk profiles were identified:

- **The "Premium Tier" Flight Risk:** The bar chart analyzing *Churn Rate by Card Category* reveals a counter-intuitive trend: **Platinum** cardholders have the highest rate of attrition (exceeding 20%), followed closely by Gold. The lowest tier, Silver, is the most stable. This suggests the bank's premium benefits are either uncompetitive or not delivering enough value to justify associated fees.
- **High-Income Attrition:** The *Churn by Card Category by Income Category* visual indicates that customers in the highest income bracket (**$120K+**) are churning at the highest rates. Interestingly, the second highest churn risk group is the lowest income bracket (**Less than $40K**), indicating a "barbell" problem where the bank is losing its wealthiest and its most vulnerable customers simultaneously.



- **Credit Behavior Clusters:** The scatter plot (*Average of Credit Limit and Sum of Total Revolving Bal by Avg Utilization Ratio*) shows a dense cluster of customers maintaining lower credit limits with varying revolving balances. A few distinct outliers show very high utilization and revolving balances, which could represent a specific segment of financially stressed customers at risk of defaulting or closing their accounts.
- **Demographic Distribution:** The pie charts reveal that a significant portion of the customer base falls into the $40K-$60K and Less than $40K income brackets, and the sum of customer ages is heavily weighted by a specific segment (making up 35.14% of the distribution).





5. Conclusion

This project successfully transitions raw banking data into a strategic business asset. By visually intersecting card categories, income levels, and credit utilization, the bank can move away from generic retention campaigns and instead create highly targeted strategies—such as reviewing the Platinum card's rewards program or offering customized credit products to the $120K+ income demographic.

6. Future Enhancements

- **Predictive Machine Learning Integration:** Integrate a Python or R script into Power BI to run a Logistic Regression or Random Forest model, assigning a specific "Churn Probability Score" (1-100%) to every active `CLIENTNUM`.
- **Financial Impact Modeling:** Add a DAX measure to calculate the exact estimated revenue lost due to the 16.07% churn rate to better quantify the ROI of retention campaigns.